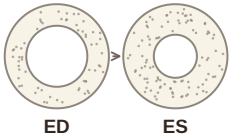
a Estimation track: cine supplies cues; correspondence is inferred; strain is defined by convention

1 · OBSERVED (I)

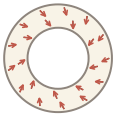
Cine bSSFP, short axis



borders conspicuous; intramural texture weak; no material label

2–3 · INFERRED (E) (R)

Estimated displacement u(X, t)



registration example: weak cues can leave the map ambiguous

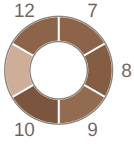
priors and supervision differ across methods

$$\hat{\varphi} = \arg\min D(I_0, I_t \circ \varphi) + \alpha R(\varphi)$$

mask agreement alone does not identify material motion (Fig. 2)

4 · DEFINED

Green–Lagrange Ecc, pixelwise



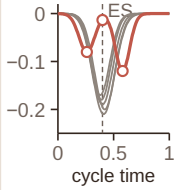
01 ED reference
02 E = ½(FᵀF − I)
03 c/r/l axes, signed
04 endo·mid·epi, 2D
05 ∂u/∂X, smoothed

fill = Ecc; seg 11 reduced

choices 01–05 fix the value before any region is reported

5 · REPORTED

Segmental Ecc(t), mid ventricle



06 AHA mid 7–12
07 at ES **−0.01**
peak sys. **−0.08**
post-sys. **−0.12**
08 min(mean) **−0.16**
mean(min) **−0.18**

choices 06–08 pick one number from the same curves (○ segment 11)

Error sources can coexist; controlled diagnostic tests:

(I) **input sufficiency** · re-image the motion more finely (E) **estimation** · vary α; retrain if required (R) **representation** · fit known field; check strain too

b Validation track: complementary endpoints; values as reported

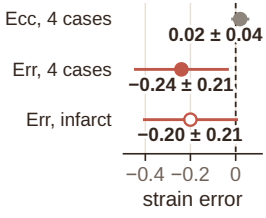
DENSE · tagging · cine FT · cine DL · ○ comparator

Known motion

MRXCAT2.0 test of DeepStrain

- ✓ technical error
- ⊗ not alone: in-vivo accuracy

estimator error (mean ± SD)



○ infarct case, EF 49%; prescribed scar Err 0.95 → 0.30 (Fig. 4)

Also Dice 0.82; displacement 1.0 ± 0.9 mm. Slice means only: no scar-region error.

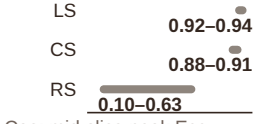
Paired DENSE / tagging

Militaru 2021; Cao 2018; Wang 2023

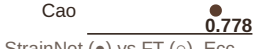
- ✓ material-sensitive agreement
- ⊗ not alone: reference error

ICC vs tagging

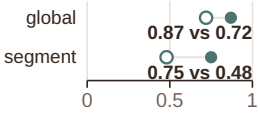
LS / CS / RS: FT–tagging



Cao: mid-slice peak Ecc



StrainNet (●) vs FT (○), Ecc



Militaru n = 61, 3 T; Cao, DENSE vs tagging; StrainNet (DL), 59 cine tests, DENSE reference.

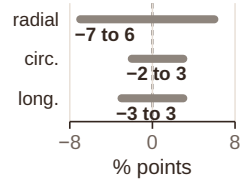
Agreement / repeatability

Yoon 2023; Bell 2025

- ✓ distinct measurement properties
- ⊗ not alone: accuracy (bias)

REGAIN vs GRAPPA

95% LoA, global FT strain



Strain-8 scan–rescan

same day, 20 volunteers
8 protocols

global	**fair–excellent**
segmental	**more variable**

Mean global differences near zero; radial limits widest (49 participants).

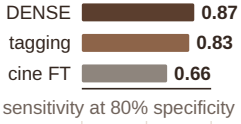
Tissue (LGE)

Kihlberg 2020

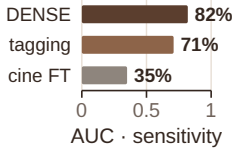
- ✓ scar-associated dysfunction
- ⊗ not alone: motion error

segments with LGE > 50%

AUC: end-systolic Ecc



sensitivity at 80% specificity



116 suspected CAD, 34 with qualifying scar. Ranks scar sensitivity, not motion accuracy.

Outcome & readers

Bello 2019; Masutani 2023

- ✓ prognostic / diagnostic link
- ⊗ not alone: clinical utility

C-index and AUC differ

Bello · RV motion in CTEPH, corrected C-index; n = 302



○ manual-variable model

Masutani · DL strain, WMA vs 4-reader consensus, AUC



DLSS: 53 patients, 846 segments. AUC spans thresholds; 30% Err is one operating point.